



Update in familial Mediterranean fever

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Purpose of review

Familial Mediterranean fever (FMF) is the prototypic autoinflammatory disease. Although the gene associated with the disease was identified 24 years ago, we still have to learn about the pathogenesis of its inflammation and the variation in the phenotype. In this review, we discuss some recent findings in FMF, such as changes in our understanding of the genetics, aims to define new criteria, and factors contributing to the disease presentation.

Recent findings

We finally have learned why a mutation causing this disease was selected in ancient times; *MEFV* gene mutations confer resistance to the microbe of plague. A group of experts have outlined recommendations for the analysis of the genetics of FMF. These recommendations complement the new classification criteria, which includes genetic results. In the past year, a number of studies have addressed the contributing factors to the inflammation caused by the mutations in pyrin; this has included epigenetic studies as well. Finally, we have long-term data for the use of anti-IL1 treatment in colchicine-resistant patients.

Summary

We now have recommendations for assessing genetic analysis of the *MEFV* gene and how to reliably classify a patient as FMF. We await further data to understand the contributing genetic and environmental factors that affect the inflammation and final phenotype in FMF and the extent of the disease presentation.

Keywords

comorbidity, familial Mediterranean fever, heterozygote, *MEFV* mutations

INTRODUCTION

Familial Mediterranean fever (FMF) is the most common autoinflammatory disease and it has opened a new chapter in rheumatology. FMF is associated with mutations in the *MEFV* gene coding for the protein pyrin. Although it is the first autoinflammatory disease to be defined, FMF continues to challenge us. In this past year, we have gained important insight into a number of new data that may solve some of our problems with this disease.

BLESSING IN DISGUISE: THE CURIOUS CASE OF THE EFFECT OF ANCIENT FAMILIAL MEDITERRANEAN FEVER MUTATIONS ON RESISTANCE TO YERSINIA PESTIS

FMF is very common in the eastern Mediterranean where the carrier rate is as high as 1/6–1/10 [1[¶]]. We have long wondered why an error, which may even cause mortality with secondary amyloidosis in the homozygous state, was selected in the region and what evolutionary force drove this change.

Six years ago a scientist from a country where FMF is very rare, introduced a milestone concept for

the activation of pyrin: they showed the Pyrin-mediated caspase 1 inflammasome activation in response to rho-glucosylation activity of *Clostridium difficile* [2]. They, therefore, suggested a new concept in our innate immune system where pyrin functions to sense pathogen modification and inactivation of Rho GTPases. This led to another groundbreaking article suggesting that the advantage of the carriers (and patients) was against *Yersinia pestis* [1[¶]] and the evolutionary pressure for selection was introduced by the microbe of the plague. The authors reach this conclusion through a sophisticated study in population genetics, and through analyzing the biology of pyrin and YopM, as well as historical records. The eastern Mediterranean was in the major crossroads between Asia and Europe in the dissemination of the

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KEY POINTS

- Carriers for MEFV mutations seem to have an advantage for an improved immune response against *Yersinia pestis*, the cause of plague.
- The presence of variants of unknown significance and other contributing factors complicate the interpretation of the genetic analysis for FMF.
- The new Eurofever/PRINTO endorsed classification criteria for FMF also includes the genetic analysis and has been validated in a multiethnic cohort.
- Anti-IL1 treatment is effective for FMF patients who are resistant to colchicine.

plague pandemic, and thus, would have been ideally located to select for FMF variants that are protective against *Yersinia* [1[¶]]. Mutations in *MEFV* were not selected globally as they were only present at the appropriate time in that limited geographic distribution [1[¶]]. The readers are referred to the elegant figures especially those presenting representative trajectories from forward-time simulation of episodic selection and how this has affected disease-causing mutations in the gene (Fig. 1d of Park *et al.* [1[¶]]).

HOW SHOULD WE ASSESS THE RESULTS OF GENOTYPING, QUO VADIS?

Sanger sequencing is still the recommended initial molecular testing for FMF. This method is well suited for low-throughput laboratories for which a next generation sequencing approach would not be profitable [3,4[¶]]. A panel of experts have developed updated recommendations for the testing of the common autoinflammatory diseases including FMF. The expert panel met face to face after two surveys and included European Molecular Genetics Quality Network members and members of ISSAID with expertise in the relevant diseases [4[¶]]. For FMF, it is recommended to screen exon 10 mutations and p.(Leu110Pro), p.(Glu148Gln), p.(Pro369Ser), p.(Arg408Gln), p.(Ile591Thr) (Table 1).

As FMF is an autosomal recessive disease, the diagnosis is confirmed if there are two pathogenic mutations. However, if one single nucleotide polymorphism is a variant of unknown significance, then diagnosis depends on clinical judgment and criteria (ISSAID/EMQN). Parental testing is recommended to resolve the issue of complex allele (cis position). On the other hand, if there is only one pathogenic mutation or two variants of unknown significance this genotype is inconclusive; the diagnosis will again rely on clinical judgment and criteria [4[¶]]. One should remember that rare variants may exist if only a limited number of exons were screened. When the genotype is inconclusive, one should consider larger testing for the other auto-inflammatory diseases as well (Fig. 1). The genetic testing is also complicated with the possible presence of digenic mutations in another *AID* gene and the rare dominant transmission as p.(Met694del) can sometimes cause an autosomal dominant disease.

In the aforementioned report, variants of unknown significance (VUS) are also defined in detail:

- (1) VUS is the first report of a gene variant associated with a typical autoinflammatory phenotype but lacking familial segregation data; thus pathogenic role is not guaranteed [4[¶]].
- (2) It is a rare/novel variant in a gene associated with recessive SAID, reported in a patient with multifactorial autoinflammatory syndrome (e.g. periodic fever, aphthous stomatitis, pharyngitis, cervical adenitis) [4[¶]].
- (3) VUS are frequent gene variants often found with various phenotypes atypical for the associated SAID and with incomplete familial segregation. Some show sub-pathogenic effect on a core pathogenic gene activity. These variants could be inflammatory or disease specific risk factors whose clinical expression depends on additional genetic/environmental factors; for example, p.(Glu148Gln) in the Mediterranean fever (*MEFV*) gene [4[¶]].

So we need good clinical criteria to define the disease, and to start treatment.

Table 1. Recommendations for screening and interpretation of variants in the *MEFV* gene

Pathogenic mutations in exon 10	Classical FMF
p.E148Q, p.L110P, p.P369S	Frequent VUS, often allelic (complex alleles)
p.P373L, p.H478Y, p.M694del, p.T577	Sometimes associated with dominant FMF-like transmission
p.P373L, p.H478Y, p.M694del, p.T577	Associated with dominant transmission (neutrophilic dermatosis)

Modified with permission from [4[¶]]. FMF, familial Mediterranean fever; VUS, variant of unknown significance.

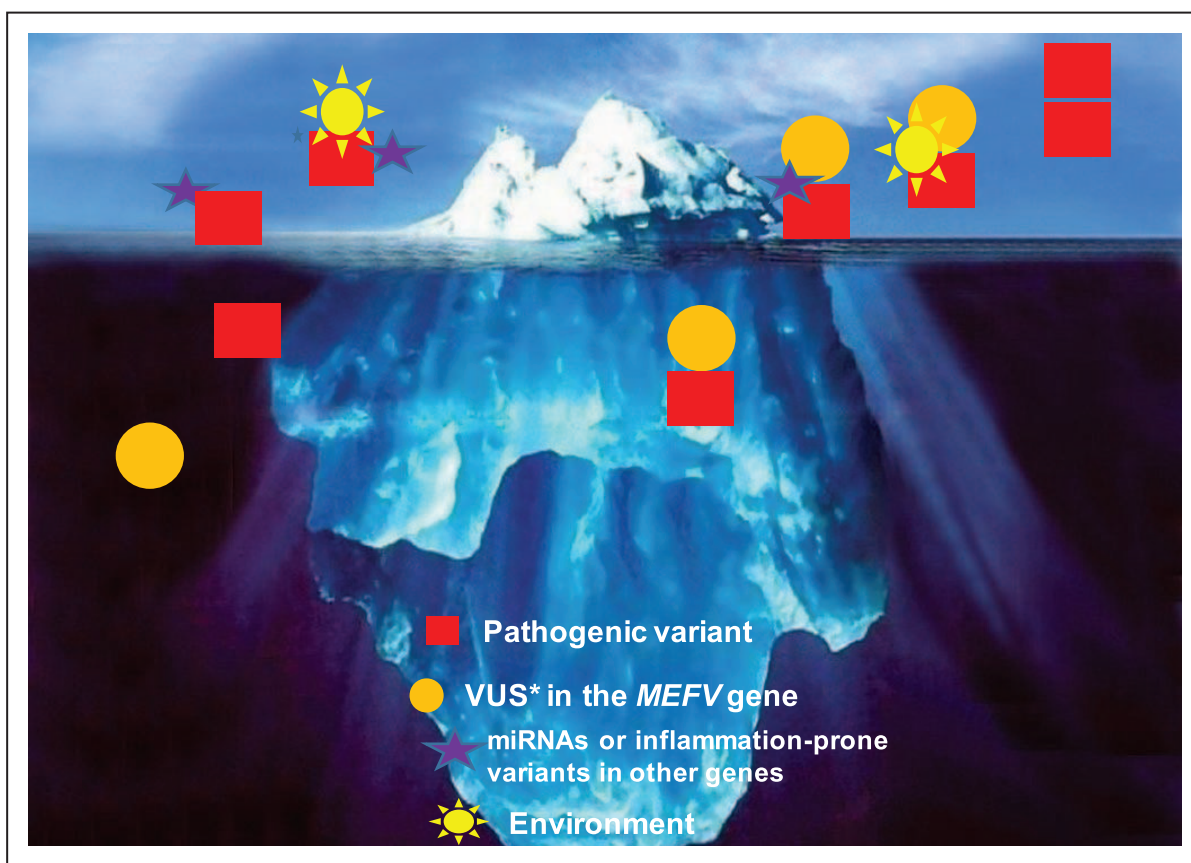


FIGURE 1. A patient with one mutation only (heterozygote) can display the familial Mediterranean fever (or familial Mediterranean fever-like phenotype) if the other allele carries single nucleotide polymorphisms that would render the individual more inflammatory, and/or has a VUS (*variant of unknown significance) as well, or because of certain triggering environmental factors.

CRITERIA, CRITERIA...

There have been previous attempts to develop classification or diagnostic criteria for FMF. However, we now have classification criteria for the four common monogenic periodic fever diseases, including the genotype result as well. These criteria were developed with a panel of 25 clinicians and 8 geneticists going through multiple steps for the selection of the best candidate classification criteria with appropriate statistical analysis; the final step was the cross-sectional validation of the novel criteria [5]. According to this new Eurofever/PRINTO endorsed criteria, in a child with confirmatory genotype for FMF, only one of the criteria below is required to classify as FMF:

- (1) Duration of episodes 1–3 days;
- (2) Abdominal pain;
- (3) Chest pain;
- (4) Arthritis.

However, if the genotype is not confirmatory (see above), you need at least two of these criteria [5]. If the genotype is not confirmatory, the author

suggests that a panel for the other common periodic fever diseases should also be done, especially if the child is not of eastern Mediterranean descent.

The performance of these criteria have already been tested in a couple of different centers. Tanatar *et al.* have compared the performance of the aforementioned criteria to the previous Tel Hashomer, Livneh and pediatric criteria. They concluded that the performances of all criteria were similar for homozygous and compound heterozygous patients. However, the new criteria did not perform so well in heterozygous patients [6].

Sag *et al.* also compared the existing criteria and concluded that the new Eurofever/PRINTO criteria had a better sensitivity but lower specificity compared with the other criteria. They suggested that the lower specificity in the Turkish cohort might have been because of high carrier rate in the population [7].

Another set of criteria addressed an unmet need: definition of colchicine resistance. The expert panel suggested definitions for both resistance and intolerance to colchicine through a series of Delphi exercises, following systematic literature review,

and a final consensus meeting [8]. This criteria defines colchicine resistance as: for a patient receiving the maximum tolerated dose of colchicine, ongoing disease activity by recurrent clinical attacks (average one or more attacks per month over a 3-month period), or persistently elevated C-reactive protein or serum amyloid A in between attacks, in the absence of any other plausible explanation. We believe these criteria are important for decision-making in the use of anti-IL1 treatment. We also believe this will guide the health authorities in defining the coverage of biologic treatment [8].

HOW DO YOU EXPRESS THE PHENOTYPE/THE DISEASE?

Patients with the FMF phenotype but with only one pathogenic mutation have been described both in areas where the disease is frequent as well as all around the world. We have long searched for answers to explain how an autosomal recessive disease could be expressed with one mutation only. A number of articles suggesting an explanation have emerged this year. Schnappauf *et al.* [3] have reported that heterozygous mutations in other domains of pyrin affect residues critical for inhibition or protein oligomerization, and lead to constitutively active inflammasome, and thus to a subclinical inflammatory phenotype in healthy carriers of FMF mutations. Indeed, we and others have shown that carriers for MEFV mutations displayed higher C-reactive protein levels and had more rheumatic or inflammatory diseases.

Umar *et al.* [9] have tried to understand how patients with one mutation expressed a disease, by using high-coverage whole genome screening. In a couple of their patients, they identified some variants in other autoinflammatory disease genes, which is the case in similar studies. They further investigated the presence of variants in other novel genes, including the exonic regions; one candidate was surprisingly in the type I interferon-signaling pathway. The second candidate gene was a variant in IL-1 receptor-like 1 gene and the deletion was present in nine FMF patients with a single mutation in the MEFV gene [9].

Akkaya-Ulum *et al.* [10] have investigated whether noncoding RNAs may have an effect in disease expression in FMF (miR197). In their FMF cohort, a number of functional assays including caspase activation suggested that a miRNA, mi-197-3p had an important role in the inflammation of these patients. They further showed through 3UTR luciferase activity assay that miR-197-3p binds to the IL-1beta receptor, *type I (IL1R1)* gene, which is critical in inflammation [10]. Karpuzoglu *et al.* [11]

have also studied miRNAs in FMF and have shown deregulation in 26 apoptosis-related miRNAs in FMF (miR). They speculated that these miRNAs may be involved in FMF pathogenesis by affecting apoptotic pathways; however, they lacked functional studies, which prevents us from drawing conclusions.

Finally the effect of microbiota has been investigated in a study from Turkey and the United States. Fecal samples of patients were collected in FMF patients with severe and mild diseases as assessed by AIDAI. The authors failed to show any difference in microbiota among severe versus mild patients in any of the regions. However, there was a difference in the fecal microbiota composition between the two countries [12]. As in the United States cohort, there were patients with high AIDAI scores as well, the authors were not able to comment on the effect of the differences between the two countries. One limitation of the study was the low number of healthy controls. The oral microbiota or other environmental factors remain to be studied.

Thus, it is tempting to suggest that carrying a MEFV mutation be a significant risk factor for symptoms or the phenotype of 'FMF'; the allele without a pathogenic mutation may contribute in a multifactorial fashion, either carrying inflammation-prone variants in the innate immune pathway or through epigenetic changes and maybe by changes introduced with the environment. The effect of VUS in the other allele may also contribute in this fashion. One may think of the FMF 'phenotype' as the portion of the iceberg above surface (Fig. 1). The portion below water is waiting to display features with the contributory factors.

IT IS NOT JUST THE ATTACKS BUT COMORBIDITIES AS WELL

As we have understood that mutations in the MEFV gene increase the inflammatory response, it is not surprising that FMF patients have associated comorbidities. Balci-Peynircioglu *et al.* [13] have analyzed the comorbidities in 2000 FMF patients; they categorized comorbidities as those associated with FMF per se, such as secondary amyloidosis; those because of increased innate inflammation and those that were regarded as incidental. The results showed that ankylosing spondylitis (sacroiliitis), Behçet disease, IgA vasculitis (Henoch Schonlein purpura), juvenile idiopathic arthritis (JIA), polyarthritis nodosa and multiple sclerosis were increased in patients with FMF [13]. The authors suggested that this was because of the increased innate immune response in carriers for the MEFV mutation. On the other hand, autoimmune diseases, such as systemic lupus erythematosus was not increased [13]. Watad *et al.*

[14] confirmed the high association of FMF with ankylosing spondylitis, Behçet disease and psoriasis; they hypothesized that the association with these MHC I-opathies suggested that the tissue-specific dysregulation of innate immunity shared between FMF and these disease may drive adaptive immune system-associated disorders [14]. However, this view does not explain the high rate of the other aforementioned diseases.

A study of 57 Chinese children with systemic onset JIA also confirmed that an exon 10 mutation in *MEFV* was a risk factor for systemic onset JIA [15]. This is in line with previous reports on systemic JIA as well.

Finally Gendelman *et al.* [16] addressed the question of whether the chronic inflammation of FMF was associated with ischemic heart disease. The multivariate analysis showed that FMF patients had an increased risk for ischemic heart disease compared with controls [16]. This was attributed to uncontrolled inflammation. As this was a large study, they were not able to assess the compliance with colchicine treatment, comorbidities or the lab values. However, this result cautions us for the need for tight inflammation control in FMF.

ANY NEW DATA IN TREATMENT?

We now have the long-term safety and efficacy data of canakinumab in colchicine-resistant FMF patients, at 113 weeks [17]. This article evaluated 60 patients who received open-label canakinumab (150 or 300 mg) in the last 72 weeks phase of the CLUSTER study. Patients had a good control of disease activity and the median C-reactive protein levels remained in the normal range. However, the serum amyloid A levels remained slightly over the normal level. No new safety signal was reported among the 57 patients who completed the study [17].

Case series for the use of anti-IL6 suggested beneficial effect in FMF. A Japanese group is now conducting an open-label continuation study of tocilizumab in FMF patients resistant or intolerant to colchicine [18]. We await the results of the efficacy and safety of anti-IL6 in FMF.

CONCLUSION

This last year has introduced us some important data, such as the evolutionary selective advantage of *MEFV* mutations as well as new criteria for classifying the disease. However, FMF still harbors many unanswered questions and continues to challenge us. We still need work to enlighten the many factors affecting the final phenotype in FMF.

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Conflicts of interest

There are no conflicts of interest.

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- of special interest
- of outstanding interest

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